



Name of the product

TECAVIR 0.5 (Entecavir Tablets 0.5 mg)/ **TECAVIR 1** (Entecavir Tablets 1 mg)

Name and strength of Active Ingredient

TECAVIR 0.5 (Entecavir Tablets 0.5 mg)

Each film-coated tablet contains: Entecavir Tablets 0.5 mg

TECAVIR 1 (Entecavir Tablets 1 mg)

Each film-coated tablet contains: Entecavir Tablets 1 mg

Dosage form

Oral dosage form, tablet

Product Description

TECAVIR 0.5 (Entecavir Tablets 0.5 mg)

White to off white, triangle shaped, biconvex film-coated tablets debossed with 'J' on one side and '110' on other side.

TECAVIR 1 (Entecavir Tablets 1 mg)

Pink, triangle shaped, biconvex, film-coated tablets, debossed with 'J' on one side and '111' on the other side.

Pharmacodynamics/Pharmacokinetics Pharmacodynamics

Pharmacotherapeutic group: antivirals for systemic use, nucleoside and nucleotide reverse transcriptase inhibitors

Mechanism of action

Entecavir, a guanosine nucleoside analogue with activity against HBV polymerase, is efficiently phosphorylated to the active triphosphate (TP) form, which has an intracellular half-life of 15 hours. By competing with the natural substrate deoxyguanosine TP, entecavir-TP functionally inhibits the 3 activities of the viral polymerase: (1) priming of the HBV polymerase, (2) reverse transcription of the negative strand DNA from the pregenomic messenger RNA, and (3) synthesis of the positive strand HBV DNA. The entecavir-TP Ki for HBV DNA polymerase is 0.0012 µM. Entecavir-TP is a weak inhibitor of cellular DNA polymerases α, β, and δ with Ki values of 18 to 40 µM. In addition, high exposures of entecavir had no relevant adverse effects on γ polymerase or mitochondrial DNA synthesis in HepG2 cells (Ki > 160 µM).

Antiviral Activity

Entecavir inhibited HBV DNA synthesis (50% reduction, EC₅₀) at a concentration of 0.004 µM in human HepG2 cells transfected with wild-type HBV. The median EC₅₀ value for entecavir against lamivudine-resistant HBV (rtL180M, rtM204V) was 0.026 µM (range 0.010–0.059 µM). Recombinant viruses encoding adefovir-resistant substitutions at either rtN236T or rtA181V remained fully susceptible to entecavir.

An analysis of the inhibitory activity of entecavir against a panel of laboratory and clinical HIV- 1 isolates using a variety of cells and assay conditions yielded EC₅₀ values ranging from 0.026 to > 10 µM; the lower EC₅₀ values were observed when decreased levels of virus were used in the assay. In cell culture, entecavir selected for an M184I substitution at micromolar concentrations, confirming inhibitory pressure at high entecavir concentrations. HIV variants containing the M184V substitution showed loss of susceptibility to entecavir.

In HBV combination assays in cell culture, abacavir, didanosine, lamivudine, stavudine, tenofovir or zidovudine were not antagonistic to the anti-HBV activity of entecavir over a wide range of concentrations. In HIV antiviral assays, entecavir at micromolar concentrations was not antagonistic to the anti-HIV activity in cell culture of these six NRTIs or entricitabine.

Resistance in cell culture:

Relative to wild-type HBV, LVDr viruses containing rtM204V and rtL180M substitutions within the reverse transcriptase exhibit 8-fold decreased susceptibility to entecavir. Incorporation of additional ETVr amino acid changes rtT184, rtS202 or rtM250 decreases entecavir susceptibility in cell culture. Substitutions observed in clinical isolates (rtT184A, C, F, G, I, L, M or S; rtS202 C, G or I; and/or rtM250I, L or V) further decreased entecavir susceptibility 16- to 741-fold relative to wild-type virus. The ETVr substitutions at residues rtT184, rtS202 and rtM250 alone have only a modest effect on entecavir susceptibility, and have not been observed in the absence of LVDr substitutions in more than 1000 patient samples sequenced. Resistance is mediated by reduced inhibitor binding to the altered HBV reverse transcriptase, and resistant HBV exhibits reduced replication capacity in cell culture.

Pharmacokinetics

Absorption and Bioavailability

It was reported that steady state was achieved after 6 to 10 days of once-daily administration with approximately 2- fold accumulation. The bioavailability of the tablet was 100% relative to the oral solution.

Effects of Food on Oral Absorption

Oral administration of 0.5 mg of entecavir with a standard high-fat meal (945 kcal, 54.6 g fat) or a light meal (379 kcal, 8.2 g fat) resulted in a delay in absorption, a decrease in Cmax of 44%–46%, and a decrease in AUC of 18%–20%

Distribution

Based on the pharmacokinetic profile of entecavir after oral dosing, the estimated apparent volume of distribution is in excess of total body water, suggesting that entecavir is extensively distributed into tissues. It was reported that binding of entecavir to human serum proteins in vitro was approximately 13%.

Metabolism

Entecavir is not a substrate, inhibitor, or inducer of the cytochrome P450 (CYP450) enzyme system.

After reaching peak concentration, entecavir plasma concentrations decreased in a bi-exponential manner with a terminal elimination half-life of approximately 128–149 hours. The observed drug accumulation index is approximately 2-fold with once-daily dosing, suggesting an effective accumulation half-life of approximately 24 hours.

Elimination

Entecavir is predominantly eliminated by the kidney with urinary recovery of unchanged drug at steady state. Renal clearance is independent of dose suggesting that entecavir undergoes both glomerular filtration and net tubular secretion.

Special Populations

Hepatic Impairment

No dosage adjustment of Entecavir is recommended for patients with hepatic impairment.

Post-liver transplant

The safety and efficacy of entecavir in liver transplant recipients are unknown.

Renal Impairment

It was reported that following a single 1-mg dose of entecavir administered 2 hours before the hemodialysis session, hemodialysis removed approximately 13% of the entecavir dose over 4 hours. CAPD removed approximately 0.3% of the dose over 7 days.

Gender

There are no significant gender differences in entecavir pharmacokinetics.

Race

There are no significant racial differences in entecavir pharmacokinetics.

Geriatric Patients

Elderly: The disparity in exposure between elderly and young subjects was most likely attributable to differences in renal function. Dosage adjustment of Entecavir should be based on the renal function of the patient, rather than age.

Pediatric Patients

Pharmacokinetic studies have not been conducted in children.

Indications

Entecavir is a nucleoside analogue indicated for the treatment of chronic hepatitis B virus infection in adults with evidence of active viral replication and either evidence of persistent elevations in serum aminotransferases (ALT or AST) or histologically active disease.

Recommended Dose

Posology

Compensated liver disease

Nucleoside naïve patients: the recommended dose in adults is 0.5 mg once daily, with or without food.

Lamivudine-refractory patients (i.e. with evidence of viraemia while on lamivudine or the presence of lamivudine resistance [LVDr] mutations): the recommended dose in adults is 1 mg once daily, which must be taken on an empty stomach (more than 2 hours before and more than 2 hours after a meal). In the presence of LVDr mutations, combination use of entecavir plus a second antiviral agent (which does not share cross-resistance with either lamivudine or entecavir) should be considered in preference to entecavir monotherapy.

Decompensated liver disease

The recommended dose for adult patients with decompensated liver disease is 1 mg once daily, which must be taken on an empty stomach (more than 2 hours before and more than 2 hours after a meal).

Duration of therapy

The optimal duration of treatment is unknown. Treatment discontinuation may be considered as follows:

- In HBeAg positive adult patients, treatment should be administered at least until 12 months after achieving HBe seroconversion (HBeAg loss and HBV DNA loss with anti-HBe detection on two consecutive serum samples at least 3-6 months apart) or until HBs seroconversion or there is loss of efficacy.
- In HBeAg negative adult patients, treatment should be administered at least until HBs seroconversion or there is evidence of loss of efficacy. With prolonged treatment for more than 2 years, regular reassessment is recommended to confirm that continuing the selected therapy remains appropriate for the patient.

In patients with decompensated liver disease or cirrhosis, treatment cessation is not recommended.

Paediatric population

For appropriate dosing in the paediatric population, Entecavir 0.5 mg film-coated tablets are available.

The decision to treat paediatric patients should be based on careful consideration of individual patient needs and with reference to current paediatric treatment guidelines including the value of baseline histological information. The benefits of long-term virologic suppression with continued therapy must be weighed against the risk of prolonged treatment, including the emergence of resistant hepatitis B virus.

Serum ALT should be persistently elevated for at least 6 months prior to treatment of paediatric patients with compensated liver disease due to HBeAg positive chronic hepatitis B; and for at least 12 months in patients with HBeAg negative disease.

Paediatric patients with body weight of at least 32.6 kg, should be administered a daily dose of one 0.5 mg tablet or 10 ml (0.5 mg) of the oral solution, with or without food. The oral solution should be used for patients with body weight less than 32.6 kg.

The recommended once-daily dose in paediatric patients weighing at least 10 kg, is presented in the table below. Patients may be dosed with or without food. Paediatric patients with body weight at least 32.6 kg, should be administered 10 ml (0.5 mg) of the oral solution or one 0.5 mg tablet once daily.

Duration of therapy for paediatric patients

The optimal duration of treatment is unknown. In accordance with current paediatric practice guidelines, treatment discontinuation may be considered as follows:

- In HBeAg positive paediatric patients, treatment should be administered for at least 12 months after achieving undetectable HBV DNA and HBeAg seroconversion (HBeAg loss and anti-HBe detection on two consecutive serum samples at least 3-6 months apart) or until HBs seroconversion or there is loss of efficacy. Serum ALT and HBV DNA levels should be followed regularly after treatment discontinuation.
- In HBeAg negative paediatric patients, treatment should be administered until HBs seroconversion or there is evidence of loss of efficacy.

Pharmacokinetics in paediatric patients with renal or hepatic impairment have not been studied.

Elderly: no dosage adjustment based on age is required. The dose should be adjusted according to the patient's renal function.

Gender and race: no dosage adjustment based on gender or race is required.

Renal impairment: the clearance of entecavir decreases with decreasing creatinine clearance. Dose adjustment is recommended for patients with creatinine clearance < 50 ml/min, including those on haemodialysis or continuous ambulatory peritoneal dialysis (CAPD). As an alternative, in case the oral solution is not available, the dose can be adjusted by increasing the dosage interval, also shown in the table. The proposed dose modifications are based on extrapolation of limited data, and their safety and effectiveness have not been clinically evaluated. Therefore, virological response should be closely monitored.

	Entecavir dosage*	
Creatinine clearance (ml/min)	Nucleoside naïve patients	Lamivudine-refractory or decompensated liver disease
≥ 50	0.5 mg once daily	1 mg once daily
30 - 49	0.25 mg once daily* OR 0.5 mg every 48 hours	0.5 mg once daily
10 - 29	0.15 mg once daily* OR 0.5 mg every 72 hours	0.3 mg once daily* OR 0.5 mg every 48 hours
< 10 Haemodialysis or CAPD**	0.05 mg once daily* OR 0.5 mg every 5-7 days	0.1 mg once daily* OR 0.5 mg every 72 hours

** on haemodialysis days, administer entecavir after haemodialysis.

Hepatic impairment: no dose adjustment is required in patients with hepatic impairment.

Mode of Administration

Entecavir tablets should be taken orally.

Contraindications

Hypersensitivity to the active substance or to any of the excipients in the formulation.

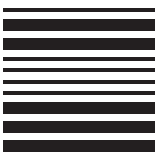
Warning and Precautions

Renal impairment: dosage adjustment is recommended for patients with renal impairment. The proposed dose modifications are based on extrapolation of limited data, and their safety and effectiveness have not been clinically evaluated. Therefore, virological response should be closely monitored.

Exacerbations of hepatitis: spontaneous exacerbations in chronic hepatitis B are relatively common and are characterised by transient increases in serum ALT. After initiating antiviral therapy, serum ALT may increase in some patients as serum HBV DNA

ARTWORK INFORMATION			
Customer	Camber	Market	Malaysia
Dimensions (mm)	240x310mm		
Printing Colours(01)	Black	Pharma code No.	F: 1924 B: 1925
Non Printing Colors	Die cut		

Note: Pharmacode position and orientation will be changed as per folding dimension



TECAVIR 0.5 / 1
Entecavir Tablets
0.5 mg and 1 mg
2081843

levels decline. Among entecavir-treated patients on-treatment exacerbations had a median time of onset of 4-5 weeks. In patients with compensated liver disease, these increases in serum ALT are generally not accompanied by an increase in serum bilirubin concentrations or hepatic decompensation. Patients with advanced liver disease or cirrhosis may be at a higher risk for hepatic decompensation following hepatitis exacerbation, and therefore should be monitored closely during therapy.

Acute exacerbation of hepatitis has also been reported in patients who have discontinued hepatitis B therapy. Post-treatment exacerbations are usually associated with rising HBV DNA, and the majority appears to be self-limited. However, severe exacerbations, including fatalities, have been reported.

Among entecavir-treated nucleoside naive patients, post-treatment exacerbations had a median time to onset of 23-24 weeks, and most were reported in HBeAg negative patients. Hepatic function should be monitored at repeated intervals with both clinical and laboratory follow-up for at least 6 months after discontinuation of hepatitis B therapy. If appropriate, resumption of hepatitis B therapy may be warranted.

Patients with decompensated liver disease: a higher rate of serious hepatic adverse events (regardless of causality) has been observed in patients with decompensated liver disease, in particular in those with Child-Turcotte-Pugh (CTP) class C disease, compared with rates in patients with compensated liver function. Also, patients with decompensated liver disease may be at higher risk for lactic acidosis and for specific renal adverse events such as hepatorenal syndrome. Therefore, clinical and laboratory parameters should be closely monitored in this patient population.

Lactic acidosis and severe hepatomegaly with steatosis: occurrences of lactic acidosis (in the absence of hypoxaemia), sometimes fatal, usually associated with severe hepatomegaly and hepatic steatosis, have been reported with the use of nucleoside analogues. As entecavir is a nucleoside analogue, this risk cannot be excluded. Treatment with nucleoside analogues should be discontinued when rapidly elevating aminotransferase levels, progressive hepatomegaly or metabolic/lactic acidosis of unknown aetiology occur. Benign digestive symptoms, such as nausea, vomiting and abdominal pain, might be indicative of lactic acidosis development. Severe cases, sometimes with fatal outcome, were associated with pancreatitis, liver failure/hepatic steatosis, renal failure and higher levels of serum lactate. Caution should be exercised when prescribing nucleoside analogues to any patient (particularly obese women) with hepatomegaly, hepatitis or other known risk factors for liver disease. These patients should be followed closely.

To differentiate between elevations in aminotransferases due to response to treatment and increases potentially related to lactic acidosis, physicians should ensure that changes in ALT are associated with improvements in other laboratory markers of chronic hepatitis B.

Resistance and specific precaution for lamivudine-refractory patients: mutations in the HBV polymerase that encode lamivudine-resistance substitutions may lead to the subsequent emergence of secondary substitutions, including those associated with entecavir associated resistance (ETVr). In a small percentage of lamivudine-refractory patients, ETVr substitutions at residues rIT184, rTS202 or rTM250 were present at baseline. Patients with lamivudine-resistant HBV are at higher risk of developing subsequent entecavir resistance than patients without lamivudine-resistance. The cumulative probability of emerging genotypic entecavir resistance after 1, 2, 3, 4 and 5 years treatment in the lamivudine-refractory studies was 6%, 15%, 36%, 47% and 51%, respectively. Virological response should be frequently monitored in the lamivudine-refractory population and appropriate resistance testing should be performed. In patients with a suboptimal virological response after 24 weeks of treatment with entecavir, a modification of treatment should be considered. When starting therapy in patients with a documented history of lamivudine-resistant HBV, combination use of entecavir plus a second antiviral agent (which does not share cross-resistance with either lamivudine or entecavir) should be considered in preference to entecavir monotherapy.

Pre-existing lamivudine-resistant HBV is associated with an increased risk for subsequent entecavir resistance regardless of the degree of liver disease; in patients with decompensated liver disease, virologic breakthrough may be associated with serious clinical complications of the underlying liver disease. Therefore, in patients with both decompensated liver disease and lamivudine-resistant HBV, combination use of entecavir plus a second antiviral agent (which does not share cross-resistance with either lamivudine or entecavir) should be considered in preference to entecavir monotherapy.

Paediatric population: A lower rate of virologic response (HBV DNA < 50 IU/ml) was observed in paediatric patients with baseline HBV DNA ≥ 8.0 log10 IU/ml. Entecavir should be used in these patients only if the potential benefit justifies the potential risk to the child (e.g. resistance). Since some paediatric patients may require long-term or even lifetime management of chronic active hepatitis B, consideration should be given to the impact of entecavir on future treatment options.

Liver transplant recipients: renal function should be carefully evaluated before and during entecavir therapy in liver transplant recipients receiving cyclosporine or tacrolimus. Co-infection with hepatitis C or D: there are no data on the efficacy of entecavir in patients co-infected with hepatitis C or D virus.

Human immunodeficiency virus (HIV)/HBV co-infected patients not receiving concomitant antiretroviral therapy: entecavir has not been evaluated in HIV/HBV co-infected patients not concurrently receiving effective HIV treatment. Emergence of HIV resistance has been observed when entecavir was used to treat chronic hepatitis B infection in patients with HIV infection not receiving highly active antiretroviral therapy (HAART). Therefore, therapy with entecavir should not be used for HIV/HBV co-infected patients who are not receiving HAART. Entecavir has not been studied as a treatment for HIV infection and is not recommended for this use. HIV/HBV co-infected patients receiving concomitant antiretroviral therapy: entecavir has been studied in 68 adults with HIV/HBV co-infection receiving a lamivudine-containing HAART regimen. No data are available on the efficacy of entecavir in HBeAg-negative patients co-infected with HIV. There are limited data on patients co-infected with HIV who have low CD4 cell counts (< 200 cells/mm³).

General: patients should be advised that therapy with entecavir has not been proven to reduce the risk of transmission of HBV and therefore appropriate precautions should still be taken.

Lactose: this medicinal product contains 120.5 mg of lactose in each 0.5 mg daily dose or 241 mg of lactose in each 1 mg daily dose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

Interaction with other Medicaments

Since entecavir is predominantly eliminated by the kidney, coadministration with medicinal products that reduce renal function or compete for active tubular secretion may increase serum concentrations of either medicinal product. Apart from lamivudine, adefovir dipivoxil and tenofovir disoproxil fumarate, the effects of coadministration of entecavir with medicinal products that are excreted renally or affect renal function have not been evaluated. Patients should be monitored closely for adverse reactions when entecavir is coadministered with such medicinal products. No pharmacokinetic interactions between entecavir and lamivudine, adefovir or tenofovir were observed.

Entecavir is not a substrate, an inducer or an inhibitor of cytochrome P450 (CYP450) enzymes. Therefore CYP450 mediated drug interactions are unlikely to occur with entecavir.

Paediatric population

Interaction studies have only been performed in adults.

Fertility Pregnancy and Lactation

Pregnancy: there are no adequate data from the use of entecavir in pregnant women. Studies in animals have shown reproductive toxicity at high doses. The potential risk for humans is unknown. Entecavir should not be used during pregnancy unless clearly necessary. There are no data on the effect of entecavir on transmission of HBV from mother to newborn infant. Therefore, appropriate interventions should be used to prevent neonatal acquisition of HBV.

Breast-feeding: it is unknown whether entecavir is excreted in human milk. Available toxicological data in animals have shown excretion of entecavir in milk. A risk to the infants cannot be excluded. Breast-feeding should be discontinued during treatment with Entecavir.

Fertility: toxicology studies in animals administered entecavir have shown no evidence of impaired fertility.

Side Effects

a. Summary of the safety profile

The most common adverse reactions of any severity with at least a possible relation to entecavir were headache, fatigue, dizziness and nausea. Exacerbations of hepatitis during and after discontinuation of entecavir therapy have also been reported.

b. Tabulated list of adverse reactions

Adverse reactions considered at least possibly related to treatment with entecavir are listed by body system organ class. Frequency is defined as very common (≥ 1/10); common (≥ 1/100 to < 1/10); uncommon (≥ 1/1,000 to < 1/100); rare (≥ 1/10,000 to < 1/1,000). Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

<i>General disorders and administration site conditions:</i>	common: fatigue
<i>Immune system disorders:</i>	rare: anaphylactoid reaction
<i>Psychiatric disorders:</i>	common: insomnia
<i>Nervous system disorders:</i>	common: headache, dizziness,
<i>Gastrointestinal disorders:</i>	common: vomiting, diarrhoea, nausea,
<i>Hepatobiliary disorders:</i>	common: increased transaminases
<i>Skin and subcutaneous tissue disorders:</i>	uncommon: rash, alopecia

Cases of lactic acidosis have been reported, often in association with hepatic decompensation, other serious medical conditions or drug exposures.

c. Paediatric Population

The adverse reactions observed in paediatric subjects who received treatment with entecavir were consistent with those observed in adults.

d. Other special populations

Experience in patients with decompensated liver disease: the safety profile of entecavir in patients with decompensated liver disease was assessed in a randomized open-label comparative study in which patients received treatment with entecavir 1 mg/day (n = 102) or adefovir dipivoxil 10 mg/day (n = 89) (study 048). Relative to the adverse reactions noted in section b. Tabulated list of adverse reactions, one additional adverse reaction [decrease in blood bicarbonate (2%)] was observed in entecavir-treated patients through week 48. The on-study cumulative death rate was 23% (23/102), and causes of death were generally liver-related, as expected in this population. The on-study cumulative rate of hepatocellular carcinoma (HCC) was 12% (12/102). Serious adverse events were generally liver-related, with an on-study cumulative frequency of 69%. Patients with high baseline CTP score were at higher risk of developing serious adverse events.

Laboratory test abnormalities: through week 48 among entecavir-treated patients with decompensated liver disease, none had ALT elevations both > 10 times ULN and > 2 times baseline, and 1% of patients had ALT elevations > 2 times baseline together with total bilirubin > 2 times ULN and > 2 times baseline. Albumin levels < 2.5 g/dl occurred in 30% of patients, lipase levels > 3 times baseline in 10% and platelets < 50,000/mm³ in 20%.

Experience in patients co-infected with HIV: the safety profile of entecavir in a limited number of HIV/HBV co-infected patients on lamivudine-containing HAART (highly active antiretroviral therapy) regimens was similar to the safety profile in monoinfected HBV patients.

Gender/age: there was no apparent difference in the safety profile of entecavir with respect to gender (≈ 25% women in the clinical trials) or age (≈ 5% of patients > 65 years of age).

Symptoms and Treatment of Overdose

There is limited experience of entecavir overdose reported in patients. Healthy subjects who received up to 20 mg/day for up to 14 days, and single doses up to 40 mg had no unexpected adverse reactions. If overdose occurs, the patient must be monitored for evidence of toxicity and given standard supportive treatment as necessary.

Effects on ability to drive and use machines

No studies on the effects on the ability to drive and use machines have been performed. Dizziness, fatigue and somnolence are common side effects which may impair the ability to drive and use machines.

Storage Condition

Store below 30°C and protect from light and moisture.

Shelf life

2 years

Dosage form and packaging supplied:

TECAVIR 0.5 (Entecavir Tablets 0.5 mg)
10x10's Alu-Alu Blister pack & 3x10's Pack

TECAVIR 1 (Entecavir Tablets 1 mg)
10x8's Alu-Alu Blister pack & 3x10's Pack

Product Registration Holder:

Camber Laboratories Sdn Bhd
Unit E-13A-02, Menara SUEZCAP 2, No.2,
KL Gateway, Jalan Kerinchi, Gerbang Kerinchi Lestari,
59200 Kuala Lumpur, Malaysia.

Manufactured by:

HETERO LABS LIMITED
UNIT-V, Sy.No: 439, 440, 441 & 458,
TSIIC Formulation SEZ, Polepally Village,
Jadcherla Mandal, Mahaboobnagar Dist.,
Telangana, INDIA.

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